Vehicle Control ECU - HLD v1

Title and Document Control

SYNTHETIC DATA FOR ACADEMIC PROJECT

Vehicle Control ECU - High Level Design

Document Version: v1

System provides coordinated vehicle-speed and engine-speed control for a synthetic passenger vehicle.

System Context

Vehicle Control ECU exchanges information with Engine Control ECU and Diagnostic Gateway. CAN is used for control traffic; Ethernet is used for diagnostic gateway uplink. Sensor Manager provides normalized vehicle and engine signals.

Software Components

VehicleControlSWC coordinates high-level vehicle behavior. EngineControlSWC manages engine-control commands. SensorManagerSWC aggregates sensor inputs. DiagnosticManagerSWC handles diagnostic requests and health reporting.

Interfaces and Ports

EngineSpeedInterface carries EngineSpeed and EngineSpeedStatus. VehicleSpeedInterface carries VehicleSpeed. EngineControlSWC provides P_EngineSpeed and P_EngineStatus and requires R_VehicleSpeed. VehicleControlSWC requires R_EngineSpeed and provides P_VehicleCommand.

Dependencies

VehicleControlSWC depends on EngineControlSWC for engine state and depends on SensorManagerSWC for vehicle speed. DiagnosticManagerSWC reads health information from EngineControlSWC and SensorManagerSWC.

Communication

CAN channel CAN0 carries EngineSpeedInterface, VehicleSpeedInterface and VehicleCommand signals. DiagnosticGatewaySWC bridges approved diagnostic traffic between CAN0 and the diagnostic Ethernet uplink.

Runnables and Signals

EngineControlSWC contains runnable UpdateEngineState. VehicleControlSWC contains runnable ComputeVehicleCommand. Signals include EngineSpeed, EngineSpeedStatus, VehicleSpeed and VehicleCommand.

Diagnostics

DiagnosticManagerSWC exposes diagnostic session handling and reads ECU health data. DiagnosticGatewaySWC forwards diagnostic requests to Vehicle Control ECU under controlled test conditions.

Architecture Rules

Every referenced interface should have a definition. Every dependency should reference a known component. Interface and port names are case-sensitive identifiers in this synthetic project.

Traceability

Key architecture elements in this document are intended for AI-assisted search, impact analysis, comparison and consistency checking. AI output requires engineering review.